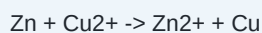


The whole idea in one chain

GOVERNED REACTION



COMPACT RULE

Track the same element -> compare oxidation state -> connect to electron loss/gain -> attach the agent role last.

WHY ORDER MATTERS

A reagent name or remembered label is not evidence. In this subtopic the self-change is the proof and the role is the conclusion.

Oxidation-state direction

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

RULE

Increase in oxidation state means oxidation. Decrease means reduction. Therefore Zn is oxidised and Cu²⁺ is reduced.

CHECK

Compare Zn with Zn and Cu with Cu. Never compare unlike elements when deciding the direction of change.

Electron consequence

ELECTRON LEDGER



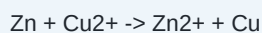
2 electrons lost = 2 electrons gained

INTERPRETATION

Zn loses two electrons; Cu²⁺ gains those two electrons. The electron ledger is an independent consistency check on the oxidation-state conclusion.

Agent roles

GOVERNED REACTION



ROLE RULE

Zn is oxidised and supplies electrons that reduce Cu^{2+} , so Zn is the reducing agent. Cu^{2+} is reduced and accepts electrons from Zn, so Cu^{2+} is the oxidising agent.

MEMORY GUARD

The reducing agent is not 'the thing that gets reduced'. Agent names describe the effect caused on the other species after the agent's own change is proved.

Quick retrieval check

TRY WITHOUT LOOKING

For $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$: write both state-change lanes, both electron statements, and both agent roles.

ANSWER

$\text{Zn } 0 \rightarrow +2$: oxidised; $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$; reducing agent. $\text{Cu } +2 \rightarrow 0$: reduced; $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$; oxidising agent.